Fanout

One driver cannot feed too many sinks without a buffer tree stage

The idea

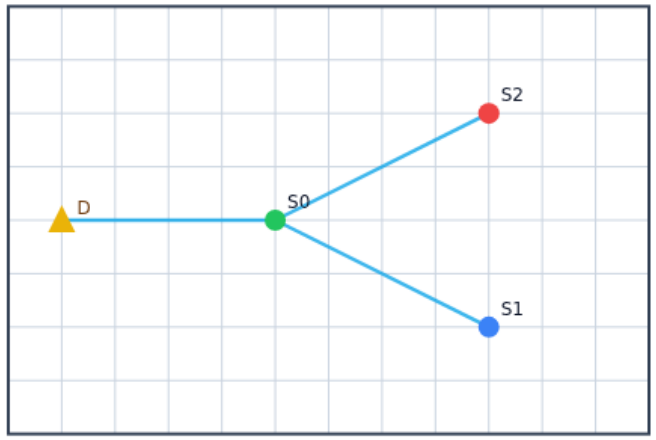
- If sink count exceeds max_fanout (two on the toy), insert BUF_X2 at S0
- Three sinks therefore force an insert

Fanout three

FANOUT TREE

Fanout three

Step 1 / 5 · fo · module03-03-fanout-tree



Explanation

Tiny net has three sinks.

- S0/S1/S2

```
sinks=3
```

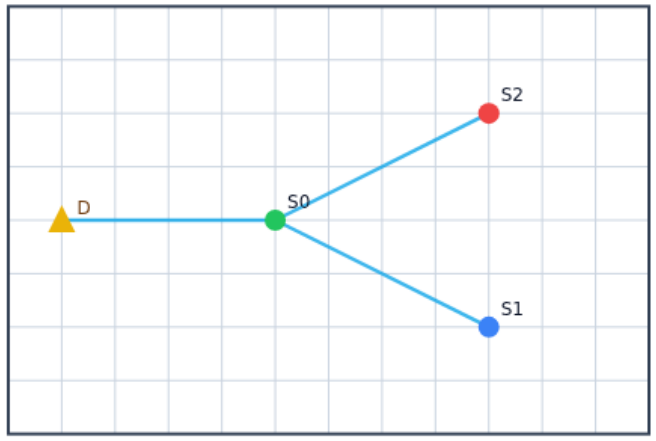
Fanout three

Max two

FANOUT TREE

Max two

Step 2 / 5 · max · module03-03-fanout-tree



Explanation

max_fanout=2 → must insert.

- Limit

```
inserted
```

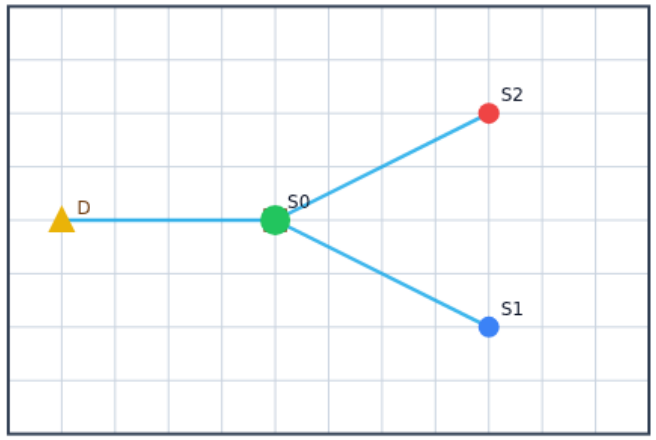
Max two

At S0

FANOUT TREE

At S0

Step 3 / 5 · s0 · module03-03-fanout-tree



Explanation

Junction buffer sits on S0.

- Steiner-ish

```
Fork
```

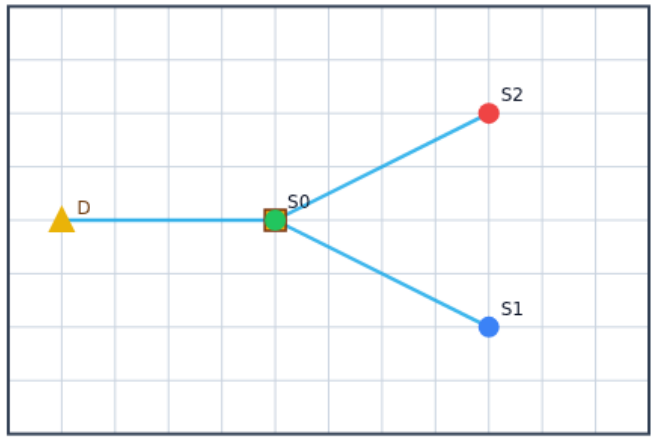
At S0

BUF_X2

FANOUT TREE

BUF_X2

Step 4 / 5 · x2 · module03-03-fanout-tree



Explanation

Stronger cell for the fork.

- Drive

```
BUF_X2
```

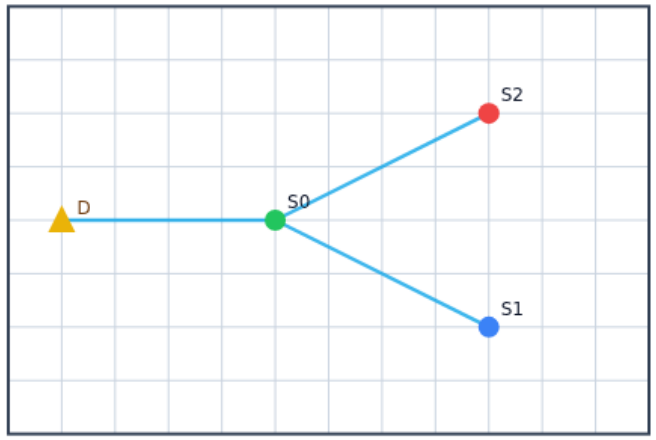
BUF_X2

Compose

FANOUT TREE

Compose

Step 5 / 5 · compose · module03-03-fanout-tree



Explanation

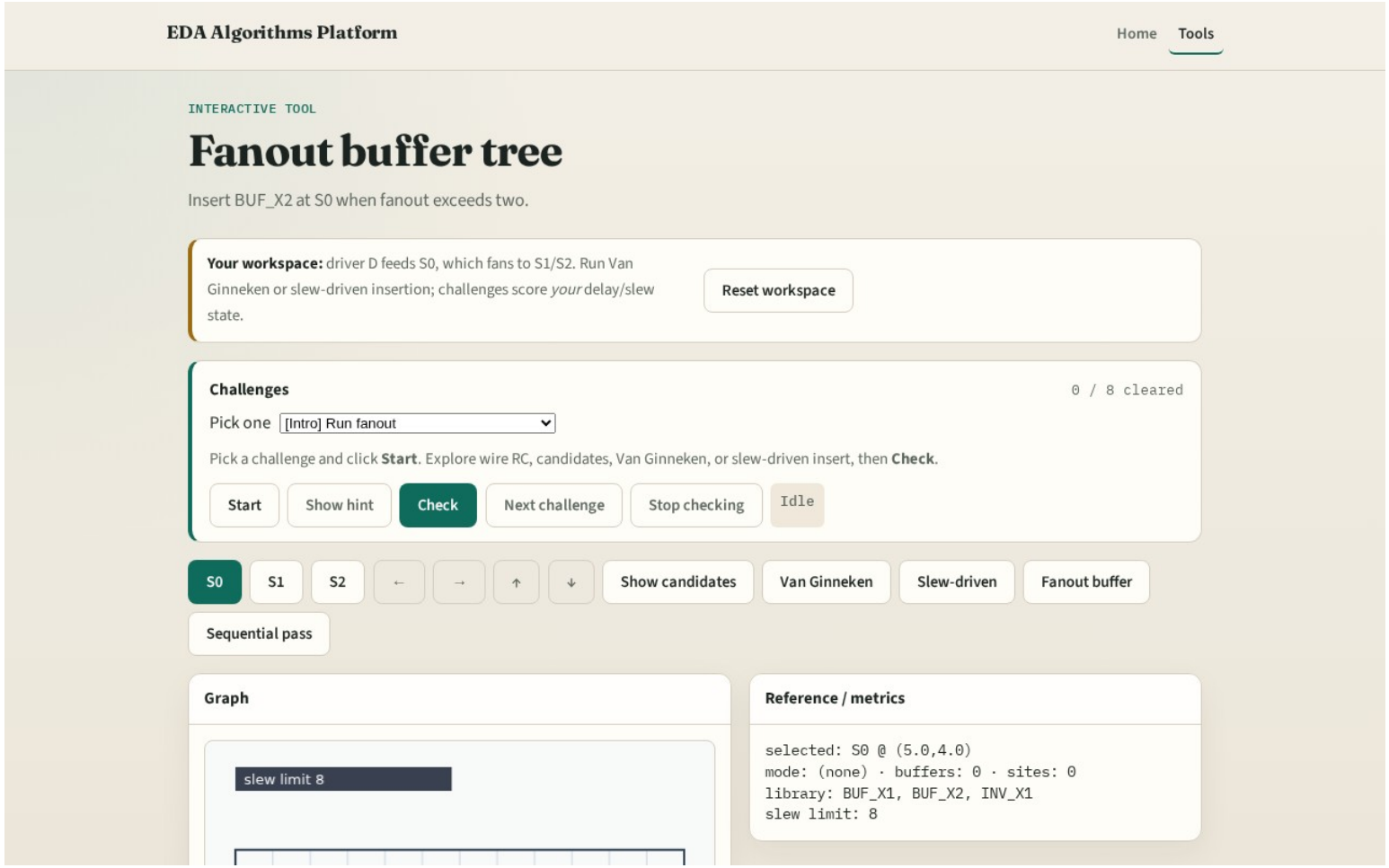
Sequential pass can combine VG + fanout.

- Pipeline

```
Next lab
```

Compose

Browser lab track



Browser lab starter

Implement track

- Call `fanout_buffer_tree(data, max_fanout=2)`
- Assert inserted is true

Pitfalls

- Buffering at the driver only
- Using max_fanout ≥ sink count
- Forgetting S0 coordinates

Your turn

- Next: full sequential buffering pass